XUGE GUO

M.S. Candidate in Mechanical Engineering | Columbia University
Robotics & Control
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Research Interests

Robot Learning · Vision-Language-Action Models · Robust Robotic Manipulation · Embodied AI
Memory-Augmented Policies · 3D Spatial Reasoning

Education

Columbia University New York, NY
M.S. in Mechanical Engineering, Focus: Robotics & Control Sep. 2025 – May. 2027 (expected)
Selected coursework: Reinforcement Learning; Model Predictive Control; Robotics Studio (A+)

Wuhan University of Science and Technology Wuhan, China
B.Eng. in Traffic Engineering June 2024

Research Experience

DynaInterfere-Bench: Robotic Manipulation under Dynamic Interference 2026 – Present
Research collaboration with Dr. Kaichen Zhou (Massachusetts Institute of Technology) and Prof. Fangneng Zhan (Hong Kong University of Science and Technology)
Controlled VLA benchmarking, baseline evaluation, and training-free recovery adaptation

- Co-developed an Isaac Sim / Isaac Lab benchmark with **six YAM Ultra2 manipulation tasks**, randomized layouts, collision-aware expert planning, and configurable visual occlusion and periodic / reactive obstacles.
- Contributed to a dataset of **2,000 successful demonstrations** (800 xArm5; 1,200 Ultra2), with dual-camera capture, image–state–action synchronization checks, train/test seed separation, and LeRobot exports.
- Integrated π0.5 / X-VLA action adapters and contributed to four-epoch, full-parameter **X-VLA fine-tuning on four RTX 5090 GPUs**, including checkpoint validation and recovery.
- Evaluated fine-tuned π0.5 and X-VLA in **400 seed-matched xArm5 trials per model**; measured **83.75% and 99.75% baseline success**, respectively, under a common 90-second task-and-return-home protocol.
- Integrated **Astra and Sol** into an RGB-based robotic control pipeline with structured Cartesian / gripper commands, inverse kinematics, and execution feedback; validated bounded motion with Astra and exploratory mug pick-and-place with Sol *without task-specific fine-tuning*.
- Prototyped a model-agnostic, training-free runtime adapter for risk monitoring, braking, avoidance, and trajectory rejoining, with an initial **12-pair on/off study (24 rollouts)**; exploring temporal memory and multi-view 3D context for robustness to occlusion.

Creative Machines Lab, Columbia University Feb. 2026 – Present
Graduate Researcher, advised by Prof. Hod Lipson

Halo Copilot / SmartHat
LLM-assisted assembly guidance with RGB-D perception and dynamic projection

- Developed a wearable camera–projector assembly assistant on **Jetson Orin Nano** with **RealSense D435i** sensing to provide spatially aligned guidance on physical workspaces.
- Combined **13-class YOLO part detection** with Lucas–Kanade optical flow and adaptive detection scheduling for continuous localization during assembly.
- Implemented depth-plane estimation, calibrated camera-to-projector mapping, and confidence-aware rendering to update object-aligned projection as viewpoints change.
- Integrated **GPT-6.1-sol** for multimodal step-completion assessment against an **eight-step XML-defined assembly plan**; investigating diagnosis of missing or incorrectly installed parts.

Research Experience (continued)

Training-Free Asynchronous Control for World-Action Models World Assembly

Execution-aware inference and asynchronous control for pretrained FastWAM policies

- Developed training-free asynchronous control extensions for FastWAM, combining continuous policy inference, execution-progress-aware action alignment, and hold-aware action stitching.
- Implemented multiprocess simulation / inference backends with dedicated GPU allocation, Pipe-based interprocess communication, reduced observation payloads, and per-stage latency profiling.
- Conducted RoboTwin baseline and ablation evaluations across **12 manipulation tasks**, with 30 rollouts per method (**720 total evaluation rollouts**).

Selected Robotics Projects

8-DOF Quadruped Robot Columbia University Robotics Studio

- Designed and built an 8-DOF quadruped using LX-16A smart servos and Raspberry Pi 4B, integrating mechanical design, inverse kinematics, gait generation, embedded actuation, and physical-robot testing.
- Developed a corresponding MuJoCo simulation and trained locomotion policies with PPO; compared learned control with manually designed gaits and investigated practical sim-to-real transfer issues.

Vision-Based Mobile Robot MakeCU Hackathon, Columbia University – 2025

Best Use of Computer Vision

- Built a two-wheel mobile robot with real-time vision-based target tracking during the MakeCU hackathon.

Teaching Experience

Columbia University Fall 2026

Teaching Assistant, MECE 4611 Robotics Studio

- Mentor student teams on robot hardware integration, servo systems, Raspberry Pi-based control, mechanical design, and system debugging; hold weekly office hours and support project evaluation.

Technical Skills

Programming: Python; Git; Linux; Bash; NumPy.

Robot learning & simulation: PyTorch; Isaac Sim; Isaac Lab; ManiSkill3; MuJoCo; reinforcement learning; PPO; OpenPI; LeRobot.

Perception: OpenCV; YOLO; RGB-D perception; optical flow; ArUco; camera calibration; RANSAC.

Robotics & hardware: Jetson Orin Nano; Raspberry Pi; Intel RealSense; smart-servo systems.

Mechanical design: SolidWorks; Fusion 360; CAD and rapid prototyping.